PREP 3, LESSONS 5–6

Internal energy and latent heat

Heating graphs; changes of state at constant temperature; $E = mL$

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Define density. DN 165 [1]

The mass per unit volume of a substance.

2 Name the method used to measure the volume of an irregularly shaped object. DN 179 [1]

The displacement method

3 A gas is heated at constant volume. State what happens to its pressure. DN 213 [1]

The pressure increases.

Key ideas LESSONS 5–6

Show your working

4 A solid is heated at a steady rate until it becomes a gas.

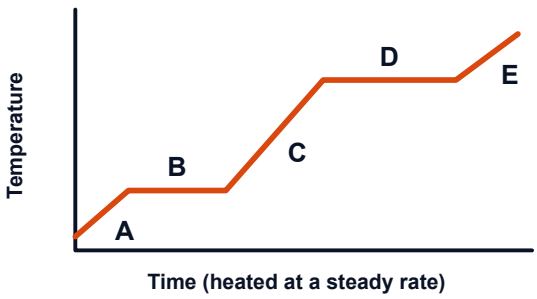
a The section that shows the substance melting is [1]

A	**B**	C	D	E

b In section C the substance is a [1]

solid	**liquid**	gas

c Explain why the temperature does not change while the substance melts. [2]



The energy supplied increases the potential energy of the particles ✓ not their kinetic energy ✓

5 Specific latent heat of vaporisation of water = 2 260 000 J/kg. Use $E = mL$

Calculate the energy needed to boil away 0.50 kg of water at 100 °C. [3]

$E = mL$ ✓

$E = 0.50 \times 2\,260\,000$ ✓

$E = 1\,130\,000$ J ✓